AGI-Oriented Reproducible Builds as Relational, Observation-Indexed GSLT Integrity

MeTTapedia Formalization Project

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Abstract

Masayuki Hatta proposes seven requirements for AGI-oriented reproducible builds, extending ordinary build reproducibility to evolving and self-modifying systems. We give a machine-checked account in which a build is a proof-relevant relation rather than necessarily a function, and reproducibility is indexed by both a declared source view and an artifact observation. The formulation separates rebuildability, per-source agreement, declared-input sufficiency, replayability, result-level hosting, proof-fibre fidelity, provenance, receipt currentness, and verification cost.

The theory proves identity and composition laws for GSLT-to-GSLT transformations, exposes hidden intermediate dependencies, and recovers functional transport only from an explicit representability certificate. It instantiates Hatta's R1–R5 as a typed profile, proves R6 for a characterized class of current protected-family-preserving self-modifications, formalizes Wheeler's diverse double-compiling assumptions and Thompson's trusting-trust attack, and treats R7 as coverage plus a Work–Span–storage feasibility discipline. Observer-indexed legibility is admitted only after exhibiting two extensionally equal realizations with different attained verification ranks. Dynamic-link provenance composes along relational routes but does not, without an additional rule, determine a legal adjudication. All positive claims are paired with negative controls, including sampled-but-not-universal verification and semantically equal routes with different provenance.

1 Attribution and scope

Hatta's seven requirements and the governance motivation come from [1, 2]. Complete input closure and purely functional deployment are informed by Dolstra and colleagues [6, 7]. The distinction between task semantics, dependency discovery, rebuilding, and scheduling follows the build-systems analysis of Mokhov, Mitchell, and Peyton Jones [8]. Signed layouts and link evidence use the conceptual separation of *in-toto* [9]. Trusting trust and diverse double-compiling are due to Thompson and Wheeler [3, 4]. The empirical and engineering reproducible-build background is provided by [5, 10].

The following are MeTTapedia extensions rather than claims attributed to those sources:

- the proof-relevant relational and observation-indexed foundation;
- the GSLT/GSLT-IL/OSLF composition and hosting theorems;
- the exact replay equivalence under stated adequacy, completeness, ordering, initial-state, and currentness premises;
- the protected-family R6 theorem for the characterized modification class;
- observer-indexed weakest-audit-view legibility and its strict separation witness;
- the R7 Work–Span–storage algebra and the dynamic-link relational provenance construction.

No theorem here converts Hatta's governance analogy into a legal conclusion. No claim about arbitrary self-modifying AGI follows from the restricted R6 result.

2 Relational build semantics

Let S be a type of complete source and environment states, A a type of artifacts, D a type of declared input views, and O a type of observable artifact outcomes. None of these types is required to be finite.

Definition 2.1 (Proof-relevant build)**.** *A build is a type-valued relation*

$$\mathcal{B} : S \to A \to \mathsf{Type}.$$

An occurrence is a triple (s, a, e) with $e : \mathcal{B}(s, a)$. Distinct evidence terms are retained even when their source and artifact endpoints coincide.

A declared input view is a map $d : S \to D$. An artifact observation is a map $o : A \to O$. Identity observation gives bit- or value-exact reproducibility; a coarser observation may express an explicitly justified tolerance or acceptance discipline.

Definition 2.2 (Separated obligations)**.** *For a build $\mathcal{B}$ and observation o:*

$$\begin{aligned}
\mathsf{Rebuildable}(\mathcal{B}) &:\Longleftrightarrow \forall s,\ \exists a,\ \mathcal{B}(s, a), \\
\mathsf{Consistent}(\mathcal{B}, o) &:\Longleftrightarrow \forall s, a, b,\ \mathcal{B}(s, a) \to \mathcal{B}(s, b) \to o(a) = o(b), \\
\mathsf{Reproducible}(\mathcal{B}, o) &:\Longleftrightarrow \mathsf{Rebuildable}(\mathcal{B}) \wedge \mathsf{Consistent}(\mathcal{B}, o).
\end{aligned}$$

Declaration sufficiency is the stronger cross-source property

$$d(s) = d(t) \wedge \mathcal{B}(s, a) \wedge \mathcal{B}(t, b) \implies o(a) = o(b).$$

Declared-view reproducibility combines rebuildability with declaration sufficiency.

This separation closes several common loopholes.

Proposition 2.3 (Vacuity and hidden input controls)**.** *The formal development proves all of the following.*

1. *Reproducibility implies rebuildability.*
2. *Rebuildability alone does not imply reproducibility: one source may admit two observably different artifacts.*
3. *Consistency alone does not imply rebuildability: an empty execution fibre is vacuously consistent.*
4. *Declaration sufficiency implies per-source consistency.*
5. *Per-source reproducibility does not imply declaration sufficiency: a hidden hardware bit can change the exact artifact while every complete source remains deterministic.*

An executable reproducer through d is stronger than fibrewise sufficiency: it computes the observed artifact from every declared value. The converse requires the exact reachability or inhabited-codomain premise needed to define the reproducer away from realized declarations. This distinction prevents a choice function outside the image from being smuggled into the definition.

2.1 Observation refinement

If an exact observation factors through a coarser observation, exact reproducibility transports downward. Transport in the opposite direction is not automatic. A numerical tolerance is accepted only through a supplied soundness law; it is not presumed transitive and therefore is not silently treated as a quotient.

The identity observation is thus the finest ordinary case, not a synonym for the general theory. This is important for AGI systems whose accepted outcome may be behavioral, probabilistic, proof-relevant, or resource-sensitive.

3 GSLT-to-GSLT composition

The core theory composes builds relationally. For $\mathcal{B}_1 : S \to M \to \mathsf{Type}$ and $\mathcal{B}_2 : M \to A \to \mathsf{Type}$, the composite evidence retains an intermediate $m : M$ and witnesses of both stages:

$$(\mathcal{B}_2 \circ \mathcal{B}_1)(s, a) := \sum_{m:M} \mathcal{B}_1(s, m) \times \mathcal{B}_2(m, a).$$

Consequently, repeated or alternative derivations are not collapsed merely because they share endpoints.

Theorem 3.1 (Identity and composition)**.** *Identity builds are reproducible. Reproducible stages compose when their declared source views include the dependency closures required by both stages and their intermediate observations are aligned. Composition is associative up to proof-fibre equivalence, rather than by erasing the intermediate witness.*

Example 3.2 (Hidden intermediate input)**.** *Two stages may each be reproducible relative to their local declarations while the composite is not reproducible relative to a declaration that hides a property read by the downstream stage. The formal counterexample keeps the middle artifact observable long enough to show exactly where sufficiency fails.*

Certified plans retain stage order, declared source demand, and observation alignment. A plan that omits a dependency is rejected even when a particular sample execution happens to return the expected endpoint.

4 Hosting, proof fibres, and open GSLT-IL routes

Result-level hosting and proof-relevant hosting are different interfaces. A behavioral hosting certificate transports statements about produced results at the declared observation. It does not imply that source and target execution fibres are equivalent. Proof-relevant hosting supplies that stronger equivalence explicitly.

Theorem 4.1 (Observation-exact transport)**.** *If a behavioral host preserves and reflects the selected artifact observation, then source reproducibility transports to the hosted build at that observation. No transport to a strictly finer observation follows without another certificate.*

The negative target called `CBad` is deterministic but invents behavior. It demonstrates why determinism and forward preservation do not establish no-invention or reflection.

GSLT-IL transport is relational by default. A relation becomes a direct function only through a proof-relevantly total and deterministic representation certificate. Open choice and partial routes remain valid relations but cannot be functionalized. Exact returned-image theorems remain scoped to the returned image; commands outside it are retained as explicit counterexamples.

The OSLF bridge follows the same discipline. Syntactic translation alone does not establish operational reproducibility. The bridge uses a named operational correspondence law, with both a satisfying instance and a failing translation witness.

5 Hatta R1–R5 as a typed profile

The formal profile stores semantic evidence, not seven Boolean flags. Its fields are summarized in Table 1.

Req.	Formal object	Required evidence
R1	declared input view	declaration sufficiency at the selected observation
R2	reproducibility budget	rebuildability and observational consistency
R3	toolchain and hardware axes	each is declared or proved observationally abstracted
R4	attestation discipline and layout	authorized, authenticated, current link for each required step
R5	source-scoped event log	initial-state, source-completeness, transition, order, and currentness premises

Table 1: The R1–R5 profile.

R3 is deliberately disjunctive: a dependency may be bound in the declared view, or a proof may show that varying it cannot affect the chosen observation. This allows a portable abstraction layer without pretending the underlying hardware or toolchain does not exist.

R4 follows the layout/link separation of *in-toto*: a layout states the required steps and authorized issuers, while a link carries occurrence-specific attestation. Authentication alone is insufficient. The negative layout accepts no issuer and therefore cannot be covered by an otherwise authentic link.

5.1 R5 and exact replay premises

Let an event system have a step relation, an authoritative initial state, an ordered authoritative event sequence, and a source-scoped retained log. The MeTTapedia replay theorem states

$$\mathsf{Trajectory}(s_0, E, s_f) \quad \Longleftrightarrow \quad \mathsf{replay}(s_0, L) = \mathsf{some}(s_f)$$

only under all of the following premises:

1. the retained and authoritative initial states agree;
2. every event occurrence retains its required sources;
3. retained events are complete as an occurrence multiset;
4. the event order is valid and recovers the authoritative sequence;
5. executable replay adequately implements the transition relation;
6. the log receipt is current at the selected revision.

Dropping a receipt reconstructs the wrong current state. Reordering a complete multiset can also fail replay. A stale log may replay successfully yet fail currentness. These controls show why replayability, source completeness, ordering, and currency remain distinct.

Prime DAG replay and HE derivation-local state-free execution instantiate this boundary separately. Their semantic objects are not collapsed into one implementation-neutral trace format.

For a state-separating open-ended query schedule, a permanently sufficient fixed context must be injective. Therefore an evolving AGI may use bounded, task-indexed views, but cannot recursively summarize away the authoritative event source and still promise sufficiency for every future separating query.

6 R6: protected recursive verifiability

R6 is represented as an indexed certificate family. The R1–R5 profile has six fibres because R3 retains separate toolchain and hardware obligations. A family-preserving modification supplies a map from every protected before fibre to its corresponding after fibre. A current modification additionally carries an issued receipt whose revision matches the current revision.

Theorem 6.1 (Restricted R6)**.** *For a current family-preserving modification, an inhabited complete R1–R5 certificate family before the modification transports to an inhabited complete family after the modification, and the modification receipt is current.*

The positive example is non-identity: it coarsens exact artifact observation to an explicit input observation, transports each semantic certificate, and keeps attestation and replay evidence. The negative candidate preserves the build relation and artifact observation but narrows the declared input view so that hardware is hidden. It cannot admit a protected-family map. A second negative uses a stale receipt.

This discharges Hatta's recursive-verifiability target for architectures whose self-modifications carry the stated family-preservation, replay, and currentness evidence. It is not an unconditional theorem about arbitrary self-modifying systems.

7 Trusting trust and diverse double-compiling

Ordinary recompilation can reproduce a malicious compiler executable from apparently benign source when the executable recognizes both the login program and its own source [3]. Matching the infected executable to its self-recompiled output is therefore not sufficient evidence.

The formal DDC model distinguishes compiler source, compiler executables, the trusted diverse compiler, source-language and executable-language semantics, the first and second stages, deterministic equality at the selected observation, and source correspondence. Wheeler's proof assumptions are represented as named fields rather than folded into a single "trust" predicate.

Theorem 7.1 (Diverse double-compiling)**.** *Under Wheeler's stated origin, compiler-correctness, determinism, trusted diverse compiler, language-compatibility, and comparison assumptions, equality of the DDC result with the compiler under test establishes correspondence to the designated compiler source at the selected observation.*

The development contains:

- a substantive benign positive instance;
- a Thompson-style infected compiler that passes ordinary self-recompilation;
- a DDC experiment that detects the infected executable;
- for every proof-one and proof-two requirement, a countermodel satisfying all other requirements while the conclusion fails;
- a control showing that matching bytes do not themselves provide the proof-one premises;
- a hosting bridge that separates result equality from proof-fibre fidelity.

The C arithmetic differential example is used only as an observation-exact hosting example. It is not mislabeled as a DDC instance because it does not construct Wheeler's complete two-stage compiler/source experiment.

8 Observer-indexed verification legibility

Legibility is not assigned to extensional behavior alone. It is indexed by an audit question, a view of the audit state, a presented realization, and an auditor capability.

Definition 8.1 (Audit information order)**.** *An audit view V is sufficient for question q when q factors through V. Write $V \preceq W$ when W refines V. Sufficient views are upward closed in this order. Two views are equivalent when each refines the other.*

Auditor capabilities enumerate permitted verification methods. A weakest-permitted method and an attained minimum rank must be supplied by proof; the theory does not take an arbitrary inconvenient certificate as the definition of opacity.

Hierarchical-complexity rank is a secondary readout. Permutation-invariant verification composition is classified as a chain; order-sensitive composition is a coordination. The same naturality square can be a chain at endpoint observation and a coordination at proof-route observation. Rank is therefore observer-relative.

Theorem 8.2 (Strict legibility separation)**.** *There exist two presented realizations with extensionally equal behavior but different attained minimum verification ranks. The lower-rank realization admits a direct evidence view; the higher-rank realization requires the full ordered trace for its audit question.*

Adding a higher-rank method cannot worsen legibility when a lower-rank method remains permitted. Conversely, low rank alone does not imply third-party attestation or DDC feasibility: both require additional authentication and compiler assumptions. This prevents a scalar clarity score from becoming a semantic authority.

9 R7: coverage and resource feasibility

R7 uses a verification plan with two independent readouts:

1. the exact multiset of checked occurrences;
2. a resource vector containing Work, Span, and retained-evidence storage.

Sequential resource composition adds Work and Span. Certified-independent parallel composition adds Work, takes the maximum Span, and retains the sum of evidence storage. Both operations are associative and monotone; the parallel readout is bounded by the sequential readout for the same components.

A verification requirement pairs a required case set with a resource limit. Meeting the requirement means both covering the set and fitting the envelope.

Theorem 9.1 (Sampling is scoped)**.** *A verification certificate proves its property at every retained checked occurrence. It proves a declared scope only with a coverage proof, and proves the universal statement only with full coverage. A one-source sample of a selectively deterministic build does not imply reproducibility at the omitted source.*

The same failure appears operationally for R5: a sampled reset receipt is valid for that event, while omission of the later increment prevents full trajectory coverage and reconstructs the wrong state.

Semantic reproducibility, current attestation, and economic feasibility are pairwise separated by concrete controls. A reproducible build can have an audit plan exceeding its budget; a cheap plan can inspect an irreproducible build; a current authentic attestation can coexist with either condition.

10 Dynamic-link provenance and plural ledgers

Hatta's protocol discussion asks composed tools to expose machine-readable license and provenance metadata for downstream reasoning [1]. The formal route type decorates an existing proof-relevant semantic relation with an ordered metadata history for every actual route witness. Each link records a link identity, tool identity, revision, source occurrences, and license occurrences.

Relational composition retains the intermediate semantic witness and concatenates the two histories. Coverage of required link sets composes under union. Repeated source and license declarations remain repeated occurrences.

Proposition 10.1 (Provenance boundaries)**.** *The development proves:*

1. *a compile–invoke chain covers both declared links;*
2. *omitting the invocation ledger entry is detectable even when the endpoint relation is unchanged;*
3. *extensionally equal routes may have different provenance histories;*
4. *attaching metadata to a nondeterministic relation does not make it functional;*
5. *compatible local ledgers glue only with an explicit gluing witness; mere runtime coordination does not manufacture one global ledger.*

Technical metadata and legal adjudication are modeled as separate fields. A factorization predicate states what would be required for artifact-plus-metadata to determine adjudication. Two cases with identical technical views and different externally supplied adjudications refute that factorization in the unconstrained model. Thus metadata enables inspection but supplies no legal rule by itself. This is a logical non-entailment result, not a legal judgment.

11 The theorem-level R1–R7 matrix

The Lean development exports a matrix whose every row contains a formal carrier, an actual positive proposition and proof, an actual negative proposition and proof, an attribution tag, and its premise boundary. Table 2 summarizes the proof terms.

Req.	Positive	Negative	Boundary
R1	complete environment declaration	input-only view hides hardware	selected observation
R2	exact identity build	exact branching build	selected budget
R3	toolchain and hardware declared	hardware neither bound nor abstracted	declare-or-abstract
R4	authorized current layout covered	rejecting layout uncovered	authorization, authentication, currency
R5	complete log reconstructs state	lost receipt breaks completeness and replay	full replay premises
R6	current non-identity coarsening transports family	behavior-only candidate and stale receipt fail	characterized modification class
R7	full two-case audit meets envelope	sample is not universal; valid build can be unaffordable	coverage plus resources

Table 2: R1–R7 theorem coverage.

The matrix is not a checklist encoding: its top theorem eliminates the seven cases and returns each row's concrete positive and negative proof terms.

12 Trust boundary and limitations

The mechanization contains no new axioms and no proof holes. Crown theorems depend only on standard Lean principles already used by the imported foundations, chiefly quotient soundness, propositional extensionality, and classical choice where representation selection requires it.

Several boundaries remain intentional.

- Reproducibility at one observation does not upgrade itself to a finer observation.
- Result equality does not imply proof-fibre identity, provenance, or trusted attestation.
- R6 applies only to current family-preserving modifications.
- Sampled verification has no universal or probabilistic conclusion without an explicit coverage or probability theorem.
- Hierarchical rank is observer- and capability-indexed; it is not a universal human-readability measure.
- Dynamic-link metadata does not determine legal status.

- A concrete CeTTa/C-subset realization is downstream work. Runtime executions may test or falsify a proposed observer, but do not define the source semantics they are meant to realize.

13 Conclusion

AGI-oriented reproducibility is most useful when treated as an invariant of a declared observation and source view, not as a synonym for deterministic or bit-identical execution. Proof-relevant relational builds expose the exact additional assumptions needed for functions, global ledgers, proof replay, and third-party trust. This makes Hatta's requirements compositional across GSLT transformations and precise enough to survive restricted self-modification, while the negative controls prevent provenance, sampling, legibility, and cost from being mistaken for semantic authority.

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